

Course: Optimization Techniques

Assignment (5+10=15 Marks): To apply optimization techniques to real-world numerical problems, demonstrating a clear understanding of the distinction between continuous (**5 marks**) and discrete (**10 Marks**) optimization problems.

1. Team Composition

- **Size:** You may work **individually** or in a **pair (max 2 students)**.
 - **Expectation:** Teams of two are expected to provide a more rigorous analysis or solve problems with higher computational complexity compared to individual submissions.
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2. Problem Selection

You must identify and define **two** distinct optimization problems:

Problem A: Continuous Optimization

Choose a problem where the decision variables X_i are real numbers ($X \in \mathbb{R}^n$).

- **Examples:** Benchmark Functions from CEC2017 can be taken (or) any research problems related to Engineering design (e.g., minimizing the weight of a bridge), chemical process control, or training a simple regression model.
- **Requirement:** The objective function must be differentiable or involve multiple decision variables.

Problem B: Discrete Optimization

Choose a problem where the decision variables are integers or involve logical choices.

- **Examples:** Teams should take distinct problems but not limited to following
 1. Traveling Salesperson Problem (TSP)
 2. Single Machine Total Tardiness Problem
 3. Flow Shop Scheduling Problem (FSSP)
 4. Quadratic Assignment Problem (QAP)
 5. Linear Ordering Problem (LOP)
 6. The Knapsack Problem (0/1)
 7. Set Cover Problem
 8. Maximum Independent Set Problem
 9. Vertex Cover Problem
 10. Maximum Clique Problem
 11. Feature Selection (in Machine Learning)

12. Portfolio Selection Problem
13. Bin Packing Problem
14. Graph Coloring Problem
15. Data Clustering
16. Number Partitioning Problem
17. Minimum Dominating set problem
18. Timetabling problem
19. Graph Partitioning problem
20. Any Binary Optimization Problems, etc.

- **Requirement:** The solution space must be finite or countably infinite, requiring combinatorial approaches.

3. Methodology

Each problem must be solved using at least **one** metaheuristic either covered during the course or any other metaheuristic of your interest. You are encouraged to compare multiple methods if applicable.

- **Methods covered in Class:** Hill climbing, Simulated Annealing, Tabu Search, ACO, GA, PSO, ABC. (You can choose other methods not covered in class also)

4. Report Requirements

Your final submission should be a structured report including: Team members details, Chosen Problems, Methods

1. **Mathematical Formulation:** Clearly define your objective functions $f(x)$, decision variables, and any equality/inequality constraints.
Use standard mathematical forms:
2. **Algorithm Implementation:** Provide a high-level explanation of the chosen algorithm and why it is suitable for the problem.
3. **Results & Convergence:** Include plots showing the objective function value over iterations (for continuous) or the progression of the best-found solution (for discrete).
4. **Discussion:** Analyze the sensitivity of the solution. Did the algorithm get stuck in local optima? How did the starting point affect the outcome?

5. Deliverables

- **Report:** A PDF document (MAX 10 pages).

- **Instances:** Either from literature or synthetic instances generated according to the problem requirements by the team
- **Code:** Well-commented source code (C / C++ / Java / Python / MATLAB) used to generate the results.

6. Important Dates

Task	Deadline
Team Registration & Problem Submission	23/04/2026
Final Submission	10/05/2026

7. First Movers Advantage for selection of problems by the teams

Before submitting chosen problems in Gform, go through the shared excel sheet as it has to be different from the previous teams which have already submitted their problem statements

ALL THE BEST